

Stage	Component	Input → Output	Key Result (ETT-h1 Case Study B)
1 DT Profiling	AAS jetson_nano (inline JSON)	Asset descriptor → SLA constraints	SLA=100 ms, RAM=3500 MB protocol=OPC-UA, sampling=1 Hz
2 Model Selection	TOPSIS (4 criteria) w_MAE=0.5, w_lat=0.3 w_par=0.1, w_lic=0.1	M_adm pool (3 models) → ranked candidates	#1 PatchTST score=1.000 #2 Lag-Llama score=0.932 #3 Moirai score=0.000
3 Quality Gate	4-check validator (latency / shape / validity / memory)	M_new=PatchTST → PASS/FAIL + fallback	FAIL: MAE==RMSE=1.303 (Phase 1 artifact) Fallback → Lag-Llama → ALL PASS
4 Drift Detection	ADWIN($\delta=0.002$) + DMCA-Drift v3.1.1 decision tree (8 feat.)	MAE stream from M_curr → detection_idx, drift_type, swap_trigger	t=12,362 (27.4% of drift window) drift_type = gradual swap_trigger = IMPROVEMENT (MAE ×0.46)
5 Re-alignment	Shadow deploy + Copilot L1 (Haiku) + COPILOTCONFIRM	M_new=Lag-Llama, SIL=1 → operator alert + atomic swap + audit JSONL	Shadow latency = 6.99 ms Copilot alert: 323 chars (auto-approve, SIL<2) Swap completed 2026-06-10T06:23:28 UTC